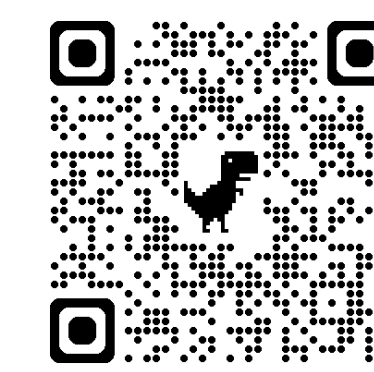
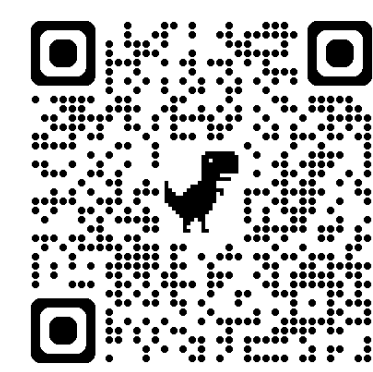


Gaussian Herding across Pens: An Optimal Transport Perspective on Global Gaussian Reduction for 3DGS

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Code



Paper

Background

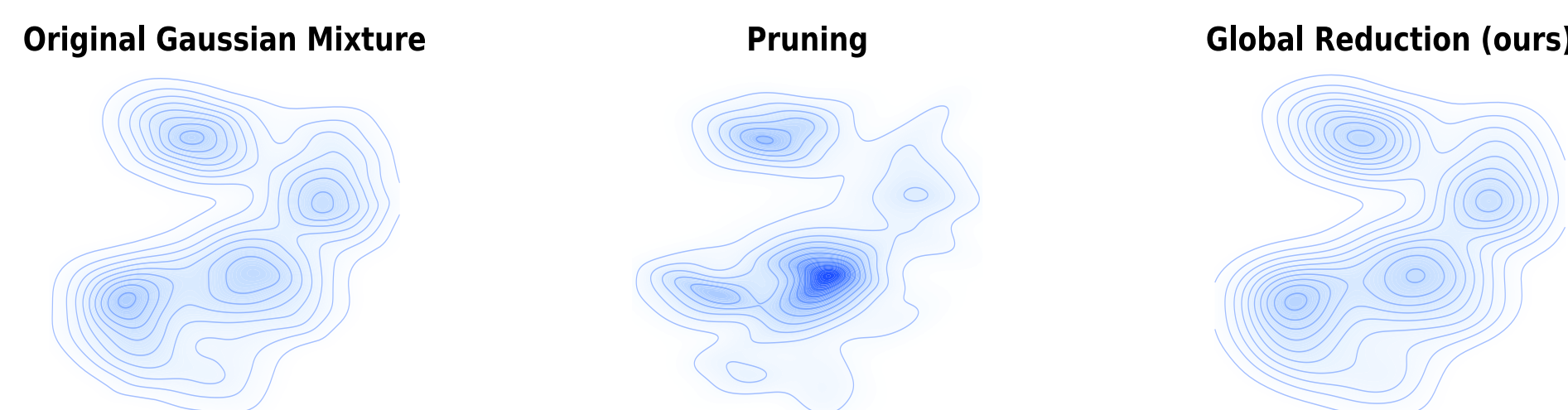
3DGS represents 3D scenes with many tiny Gaussian primitives, enabling fast, photorealistic rendering of complex environments.

However, 3DGS often contains **millions of redundant** Gaussians, which increase memory usage and slow down rendering.

One way to address this is through compaction—reducing the number of Gaussians while preserving visual quality.

- Pruning based** (Compacting by removal with low score): **Light-GS** (Fan et al. 2024), **PUP-3DGS** (Hanson et al. 2024), **Tramming-the-fat** (Ali et al. 2024)
- End-to-end** (Directly training a lighter-weight 3DGS model) : **3DGS-MCMC** (Kheradmand et al. 2024), **LocoGS** (Shin et al. 2025)

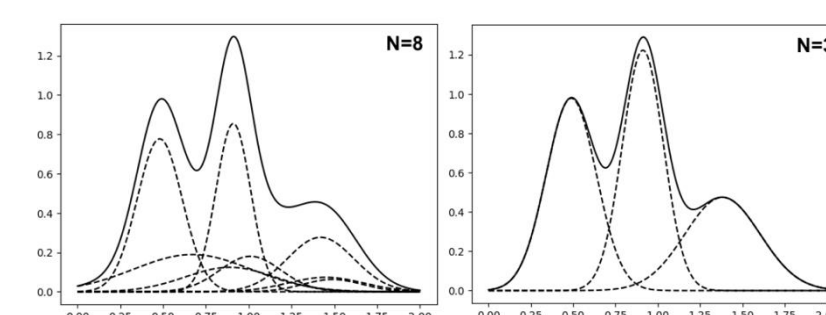
Pruning may break geometric consistency, so we propose a **global** approach that **preserves both shape and appearance**.



Gaussian mixture reduction

We interpret the 3DGS model as a **Gaussian Mixture Model**:

$$\phi_n(x) = \sum_{i=1}^n \alpha_i \phi(x; \mu_i, \Sigma_i),$$



Example of GMR. The right one has same overall shape as left, but with fewer Gaussians

The compacted 3DGS model is also a Gaussian mixture:

$$\phi'_m(x) = \sum_{j=1}^m \alpha'_j \phi(x; \mu'_j, \Sigma'_j)$$

Find $\phi'_m(x)$ that minimizes $\mathcal{T}_c(\phi_n, \phi'_m)$

$$\mathcal{T}_c(\phi_n, \phi'_m) = \inf \left\{ \sum_{i=1}^n \sum_{j=1}^m \pi_{ij} c(\phi(\cdot; \mu_i, \Sigma_i), \phi(\cdot; \mu'_j, \Sigma'_j)) : \sum_{j=1}^m \pi_{ij} = \alpha_i, \sum_{i=1}^n \pi_{ij} = \alpha'_j \right\}$$

$$\{\bar{\alpha}_j, \bar{\mu}_j, \bar{\Sigma}_j\} = \arg \min_{\{\alpha'_j, \mu'_j, \Sigma'_j\}} \mathcal{T}_c(\phi_n, \phi'_m)$$

Why composite transportation divergence?

Efficient k-means-like algorithm: group similar Gaussians by proximity to current Gaussian centers, then update the centers.

For the cost below, cluster centers have closed-form updates.

$$c(\phi, \phi') = \|\mu - \bar{\mu}\|_2^2 + \|\Sigma - \bar{\Sigma}\|_F^2, \quad \bar{\mu}_j = \frac{\sum_{i \in \mathcal{C}_j} \alpha_i \mu_i}{\sum_{i \in \mathcal{C}_j} \alpha_i}, \quad \bar{\Sigma}_j = \frac{\sum_{i \in \mathcal{C}_j} \alpha_i \Sigma_i}{\sum_{i \in \mathcal{C}_j} \alpha_i}.$$

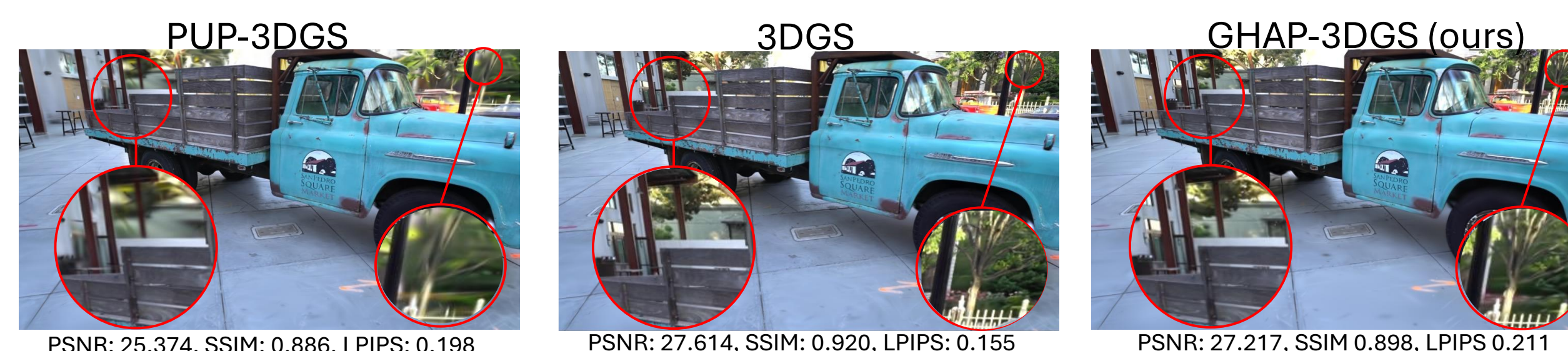
Highlight

A global, principled, and optimal transport based 3DGS compaction method.

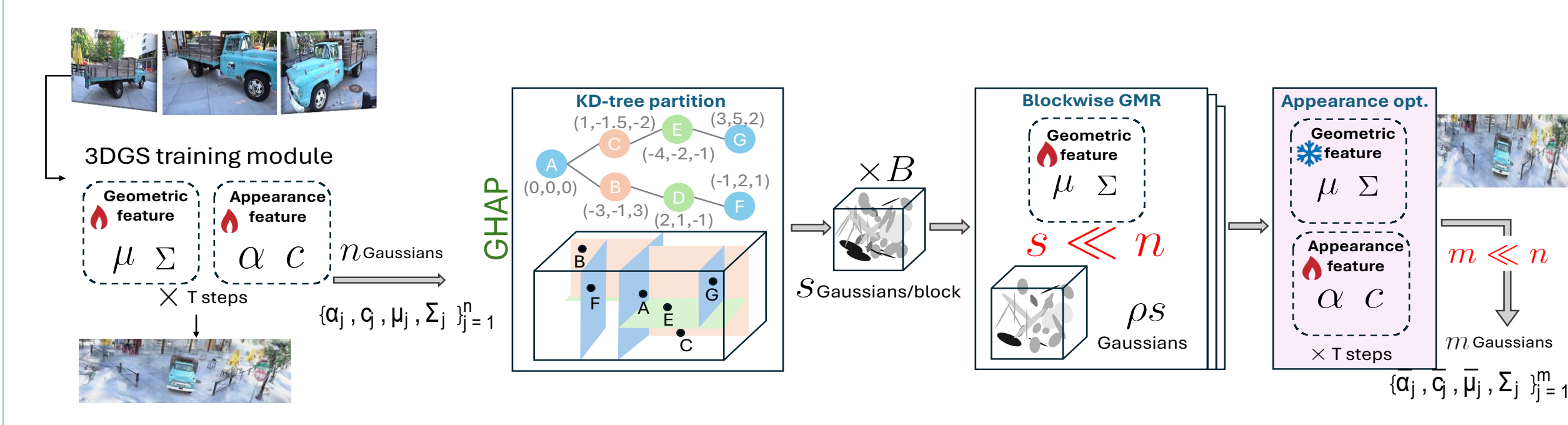
Reinterprets “compaction” as **global Gaussian Mixture Reduction (GMR)**.

GMR doesn’t just prune — it rethinks the representation, yielding **fewer but smarter Gaussians**.

Cuts Gaussian counts by **up to 90%** while preserving rendering quality.



Full pipeline



Summary

We develop a new 3DGS compaction framework that effectively preserves the geometric fidelity of the original model.

We design an efficient pipeline for high-precision compaction of large-scale 3DGS models.

Our method is post-hoc and compatible with any existing 3DGS pipeline, making it highly practical and broadly applicable.

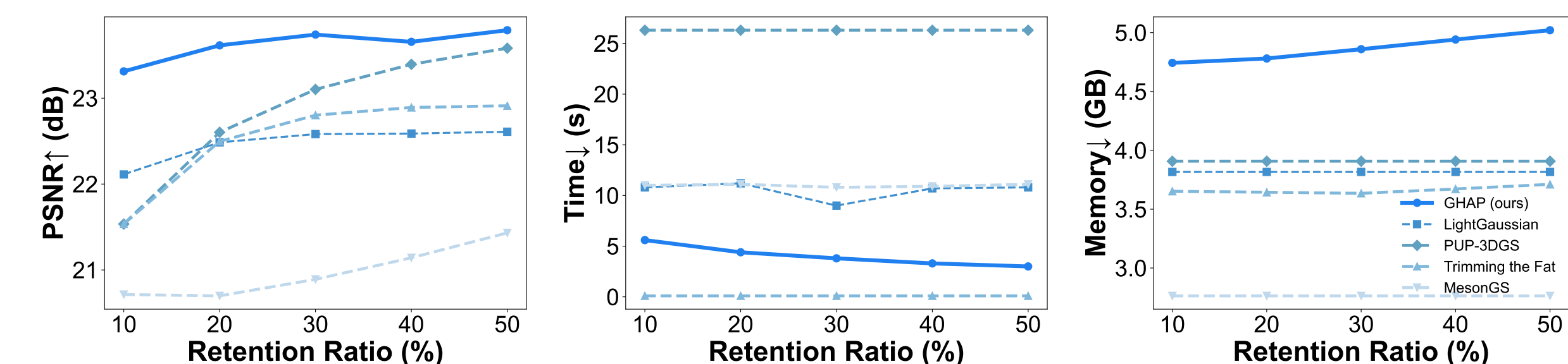
Experiments

Comparison with SOTA

Method	Tanks&Temples				MipNeRF-360				Deep Blending			
	SSIM↑	PSNR↑	LPIPS↓	k Gaussians	SSIM↑	PSNR↑	LPIPS↓	k Gaussians	SSIM↑	PSNR↑	LPIPS↓	k Gaussians
Vanilla 3DGS	0.853	23.785	0.169	1577	0.813	27.554	0.221	2627	0.907	29.816	0.238	2475
LocoGS	0.843	23.655	0.191	571	0.798	27.049	0.257	674	0.903	29.972	0.261	529
3DGS+GHAP (ours)	0.818	23.312	0.242	157	0.764	26.404	0.314	263	0.905	29.647	0.264	248
LightGaussian	0.756	22.113	0.306	158	0.735	25.674	0.331	263	0.869	28.010	0.327	248
PUP-3DGS	0.767	21.519	0.280	158	0.753	25.332	0.309	262	0.895	29.153	0.274	248
Trimming the Fat	0.776	21.535	0.293	156	0.731	25.255	0.343	263	0.887	28.056	0.302	247
MesonGS	0.811	20.714	0.208	157	0.773	24.924	0.264	263	0.896	28.693	0.264	248
3DGS-MCMC	0.779	22.141	0.282	157	0.763	25.957	0.309	263	0.885	28.976	0.298	248
MiniSplatting	0.799	22.661	0.265	78	0.759	26.022	0.318	111	0.895	29.395	0.289	125
MiniSplatting+GHAP (ours)	0.835	23.232	0.198	79	0.802	27.090	0.250	112	0.909	30.042	0.254	127

Our method (GHAP) consistently matches or surpasses pruning-based and end-to-end baselines, even at a low retention rate (10%)

RD curve, runtime & memory comparison



GHAP outperforms pruning-based approaches across different retention levels, achieving faster compaction with a slight increase in memory usage.

References

- Fan et al. *NeurIPS* 2024
- Hanson et al. *CVPR* 2024
- Ali et al. *BMVC* 2024
- Kheradmand et al. *NeurIPS* 2024
- Shin et al. *ICLR* 2025
- Zhang et al. *TIT* 2024